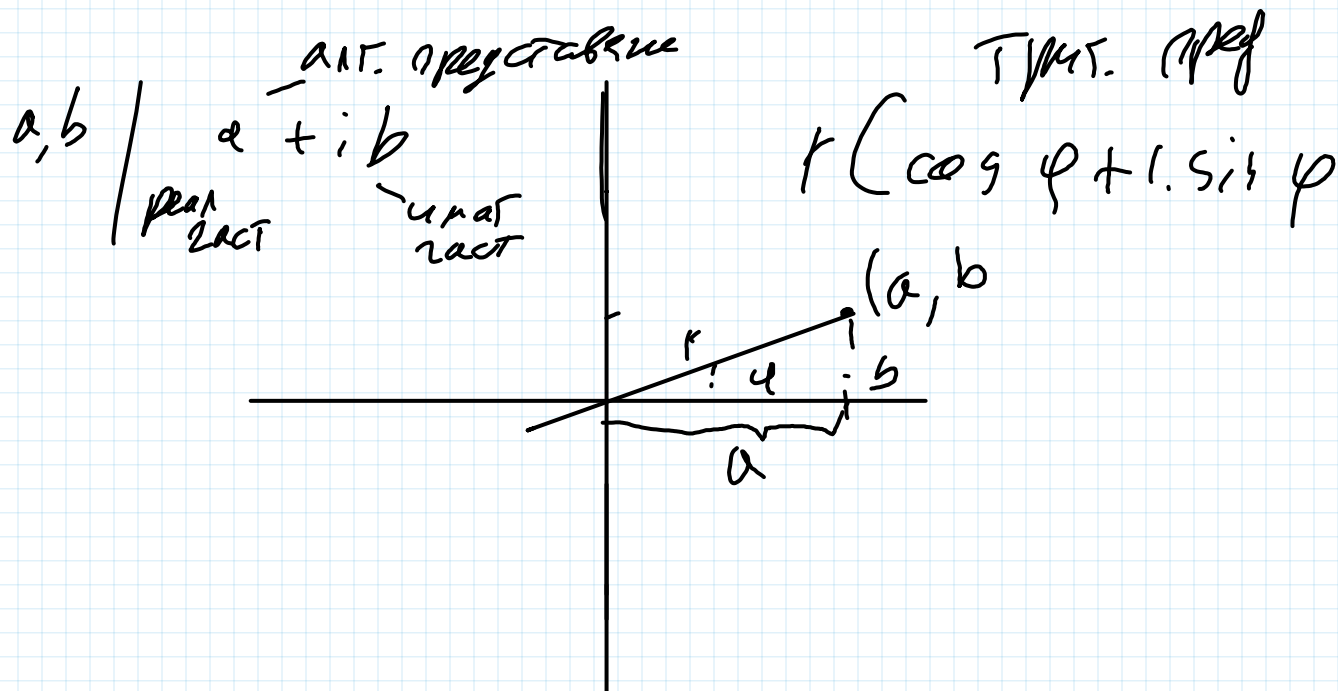


Комплексные числа



$$+ (a_1 + ib_1) + (a_2 + ib_2) = a_1 + a_2 + i(b_1 + b_2)$$

$$- (a_1 + ib_1) - (a_2 + ib_2) = a_1 - a_2 + i(b_1 - b_2)$$

$$\times (a_1 + ib_1)(a_2 + ib_2) = a_1 a_2 + i^2 b_1 b_2 + i(a_1 b_2 + a_2 b_1)$$

$$\div \frac{(a_1 + ib_1)(a_2 - ib_2)}{(a_2 + ib_2)(a_2 - ib_2)} = \frac{a_1 a_2 - i^2 b_1 b_2 + i(a_1 b_2 - a_2 b_1)}{a_2^2 - i^2 b_2^2} = \frac{a_1 a_2 + b_1 b_2 + i(a_1 b_2 - a_2 b_1)}{a_2^2 + b_2^2} = \frac{\dots}{a_2^2 + b_2^2} = \frac{\dots}{r^2}$$

$$1) a) (2 + 5i) + (3 - 4i) = 5 + i(5 - 4) = 5 + i$$

$$b) (2 + 5i) - (3 - 4i) = -1 + 9i$$

$$c) (2 + 5i)(3 - 4i) = 6 - 20i^2 + i(-8 + 15) = 6 + 20 + 7i = 26 + 7i$$

$$d) \frac{2 + 5i}{3 - 4i} = \frac{2 + 5i}{3 - 4i} \cdot \frac{3 + 4i}{3 + 4i} = \frac{14 + 23i}{25}$$

$$2 \quad a) 5 \left(\cos \frac{\pi}{8} + i \sin \frac{\pi}{8} \right)^3 \left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right) = 15 \left(\cos \left(\frac{\pi}{8} + \frac{3\pi}{2} \right) + i \sin \left(\frac{\pi}{8} + \frac{3\pi}{2} \right) \right) \\ = 15 \cdot \left(\cos \frac{13\pi}{8} + i \sin \frac{13\pi}{8} \right)$$

$$r_1 (\cos \varphi_1 + i \sin \varphi_1) \cdot r_2 (\cos \varphi_2 + i \sin \varphi_2) = r_1 \cdot r_2 (\cos(\varphi_1 + \varphi_2) + i \sin(\varphi_1 + \varphi_2))$$

$$\sin(x+y) = \sin x \cos y + \sin y \cos x$$

$$\sin(x-y) = \sin x \cos y - \sin y \cos x$$

$$\cos(x+y) = \cos x \cos y - \sin x \sin y$$

$$\cos(x-y) = \cos x \cos y + \sin x \sin y$$

$$b) \frac{5 \left(\cos \frac{\pi}{8} + i \sin \frac{\pi}{8} \right)}{3 \left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right)} = \frac{5}{3} \cdot \left(\cos \left(\frac{\pi}{8} - \frac{3\pi}{2} \right) + i \sin \left(-\frac{11\pi}{8} \right) \right) = \\ = \frac{5}{3} \cdot \left(\cos \frac{-11\pi}{8} + i \sin \frac{-11\pi}{8} \right)$$

$$* z = a + ib$$

$$\bar{z} = a - ib$$

конjugат на z

$$\frac{r_1 (\cos \varphi_1 + i \sin \varphi_1)}{r_2 (\cos \varphi_2 + i \sin \varphi_2)} = \frac{r_1}{r_2} \cdot (\cos(\varphi_1 - \varphi_2) + i \sin(\varphi_1 - \varphi_2))$$

$$③ (1+i)^5 - 3i - \text{Дом} \quad (r (\cos \varphi + i \sin \varphi))^n = r^n (\cos(n\varphi) + i \sin(n\varphi)) \\ 5_1 (2i)^{283} =$$

* 1 шаг - сложение

$$2 \text{ шаг} \text{ биномнa фopм } (a+b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

3 шаг - триг.

*

0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
10 cos 90°	$\frac{\sqrt{3}}{2}, \frac{1}{2}$	$\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}$	$\frac{1}{2}, \frac{\sqrt{3}}{2}$	01

$$\frac{(1+i)^3 - (1-i)^4}{(4-2i)(3-2i)} = \frac{11i-3}{65} \quad - \text{Done}$$

$$\ast \quad x^n = z \quad x = ? \quad x^n = r(\cos \varphi + i \sin \varphi) \quad x = \sqrt[n]{r} \left(\cos \frac{\varphi + 2k\pi}{n} + i \sin \frac{\varphi + 2k\pi}{n} \right)$$

⑤ $x^4 = 1$ $x = ?$

$\vec{p} \cdot \vec{e} = 1 = \cos 0 + i \sin 0$

$$\Rightarrow x = \sqrt[n]{1} \left(\cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n} \right) = \cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n} = \left(\cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n} \right)^k, \quad k \in \mathbb{R}$$

$$\int_1^7 x^3 = 7 \rightarrow x = \cos \frac{2\pi i}{3} + i \sin \frac{2\pi i}{3}$$

$$k=0 \Rightarrow x_0 = \cos 0 + i \sin 0 = 1$$

$$k_1 \Rightarrow x_1 = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} = -\frac{1}{2} + i \frac{\sqrt{3}}{2}$$

$$k_2 \Rightarrow x_2 = \cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} = -\frac{1}{2} - i \frac{\sqrt{3}}{2}$$

ако α е корен и комплексно
срещното му е корен

$$⑥ x^2 + 6x + 13 = 0$$

$$D = b^2 - 4ac$$

$$D = -4$$

$$D < 0$$

$$x = ?$$

$$x_{1,2} = \frac{-b \pm \sqrt{D}}{2a}$$

$$x_{1,2} = \frac{-6 \pm \sqrt{-4}}{2}$$

$$D = -4$$

$$x_{1,2} = \frac{-6 \pm \sqrt{-4}}{2} = \frac{-6 \pm 2i}{2} = -3 \pm i$$

$$\text{Заг } a) x^2 = 3 - 4i \quad x = ?$$

$$x = a + ib \quad a, b \in \mathbb{R}$$

$$(a + ib)^2 = 3 - 4i$$

$$a^2 + 2iab + i^2 b^2 = 3 - 4i$$

$$a^2 + 2iab - b^2 = 3 - 4i$$

$$\begin{cases} a^2 - b^2 = 3 \\ 2ab = -4 \end{cases} \quad \begin{cases} a = \pm 2 \\ b = \mp 1 \end{cases}$$

$$x = \pm(2 - i)$$

$$x^2 - x + 1 - i = 0$$

$$D = 1 - 4(1 - i) = 1 - 4 + 4i = -3 + 4i$$

$$\sqrt{D} = ? \quad \sqrt{D} = \sqrt{-3 + 4i} =$$

$$= \sqrt{(1 - 2i)(1 + 2i)} =$$

$$= \sqrt{1 - 4i^2} =$$

$$= \sqrt{1 + 4} = \sqrt{5}$$

$$= \pm(1 + 2i)$$

$$x_{1,2} = \frac{1 \pm (1 + 2i)}{2} \quad x_1 = \frac{1 + 1 + 2i}{2} = 1 + i$$

$$x_2 = -i$$

$$2x - |x| = 3 - 6i$$

P-u $x = a + bi$

$$|x| = \sqrt{a^2 + b^2}$$

$$2a + 2bi - \sqrt{a^2 + b^2} = 3 - 6i$$

$$\begin{cases} 2a - \sqrt{a^2 + b^2} = 3 \\ 2b = -6, b = -3 \end{cases}$$

$$1 = 1 + 0i \quad |1| = \sqrt{1^2 + 0^2} = 1$$

$$\sqrt[n]{1} = \sqrt[n]{1(\cos 0 + i \sin 0)}$$

$$= \sqrt[n]{1} \left(\cos \frac{0 + 2k\pi}{n} + i \sin \frac{0 + 2k\pi}{n} \right) =$$

$$= \cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n}, k = 0, 1, \dots, n-1$$

$$\sqrt{a^2 + b^2} = 2a - 3 \quad |^2$$

$$a^2 + b^2 = 4a^2 - 12a + 9$$

$$3a^2 - 12a = 0$$

$$a(3a - 12) = 0$$

$$a = 4, 0$$

CAY Метод на Гаусс

систем линейных уравнений

n уравнений

$$3x = 1 \text{ или } y = e$$

$$3x + 5y = 5$$

$$\begin{cases} 3x + 5y = -1 \\ y = 2 \end{cases} \quad \text{C-1a}$$

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_n \end{cases}$$

$$\left(\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & & a_{2n} & b_2 \\ \vdots & & & & \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array} \right)$$

ЕЛ-преобразован (не променя р-и т.)

1) Ръководство на гръ

2) $y_{ii} \neq 0$

3) Добавяне към гр-е да гр гр-е е изключено с мидо

$$\begin{array}{l} 1 \mid x_1 - x_2 + x_3 + x_4 = 1 \\ 3x_1 + x_2 - x_4 = 2 \end{array}$$

$$\left(\begin{array}{cccc|c} 1 & -1 & 1 & 1 & 1 \\ 3 & 1 & 0 & -1 & 2 \end{array} \right) \quad R_2 - 3R_1$$

$$\left(\begin{array}{cccc|c} 1 & -1 & 1 & 1 & 1 \\ 0 & 4 & -3 & -4 & -1 \end{array} \right) \quad R_2 \cdot \frac{1}{4} \quad \begin{array}{l} 4x_2 - 3x_3 - 4x_4 = -1 \\ x_2 = \frac{-1 + 3x_3 + 4x_4}{4} \end{array}$$

Полагам $x_3 = p, \quad x_4 = q \Rightarrow x_2 = \frac{-1 + 3p + 4q}{4}$

$R_1: x_1 - x_2 + x_3 + x_4 = 1$

$$x_1 = 1 + x_2 - x_3 - x_4 = 1 + \frac{-1 + 3p + 4q}{4} - p - q =$$

$$\frac{4 - 1 + 3p + 4q - 4p - 4q}{4} = \frac{3 - p}{4}$$

$$(x_1, x_2, x_3, x_4) =$$

$$\left(\frac{3-p}{4}, \frac{-1+3p+4q}{4}, p, q \right)$$

$$7 \text{ б) } \begin{cases} x_1 - x_2 + 5x_3 = 1 \\ 3x_1 - 8x_2 + 16x_3 - x_4 = 4 \\ x_3 + x_4 = 8 \end{cases}$$

$$\text{б) } \left(\begin{array}{cccc|c} 1 & -1 & 5 & 0 & 1 \\ 3 & -8 & 16 & -1 & 4 \\ 0 & 0 & 1 & 1 & 8 \end{array} \right) \quad R_2 - 3R_1$$

$$\left(\begin{array}{cccc|c} 1 & -1 & 5 & 0 & 1 \\ 0 & -5 & 1 & -1 & 1 \\ 0 & 0 & 1 & 1 & 8 \end{array} \right)$$

$$x_4 = p$$

$$x_3 = 8 - p$$

$$\begin{aligned} -5x_2 &= 1 - x_3 - x_4 = \\ &= 1 - 8 - p + p = \\ x_2 &= (-7 + 2p) / (-5) \end{aligned}$$

$$x_1 = 4x_2 - 5x_3 = 1 - \frac{-7+2p}{5} - 5(8-p) =$$

$$= \frac{5+7-2p-540+25p}{5} = \quad x_1 = \frac{-188+23p}{5}$$

$$\left(\frac{-188+23p}{5}, \frac{7-2p}{5}, 8-p, p \right)$$

p — произвольное

$$\left(\begin{array}{ccccc|c} 1 & 1 & -1 & 1 & -1 & 1 \\ 2 & 1 & 0 & 1 & 0 & 7 \\ 1 & 0 & -1 & 0 & -1 & 4 \\ -1 & 1 & 0 & 2 & 1 & -3 \\ 0 & 0 & 0 & 1 & 2 & 2 \end{array} \right)$$

OSHO

$$\left(\begin{array}{ccccc|c} 1 & 1 & -1 & 1 & -1 & 1 \\ 0 & -1 & 2 & -1 & 2 & -1 \\ 0 & -1 & 0 & -1 & 0 & 3 \\ 0 & 2 & -1 & 3 & 0 & -2 \\ 0 & 0 & 0 & 1 & 2 & 2 \end{array} \right)$$

$$\left(\begin{array}{ccccc|c} 1 & 1 & -1 & 1 & -1 & 1 \\ 0 & -1 & 2 & -1 & 2 & -1 \\ 0 & 0 & 0 & 1 & 0 & 4 \\ 0 & 2 & -1 & 3 & 0 & -2 \\ 0 & 0 & 0 & 1 & 2 & 2 \end{array} \right) \quad \left(\begin{array}{ccccc|c} 1 & 1 & -1 & 1 & -1 & 1 \\ 0 & -1 & 2 & -1 & 2 & -1 \\ 0 & 0 & 0 & 1 & 0 & 4 \\ -1 & 1 & 0 & 2 & 1 & -3 \\ 0 & 0 & 0 & 1 & 2 & 2 \end{array} \right)$$

$$x_4 = 4$$

$$x_5 = -1$$

$$-x_2 + 2x_3 - 4 - 2 = -1$$

$$-x_2 + 2x_3 = 5$$

$$-x_1 + x_2 + 8 - 1 = 3$$

$$-x_1 + x_2 = -10$$

$$x_2 = 2x_3 - 5 = x_3 - 10$$

$$x_1 + x_2 - x_3 + 4 + 1 = 1$$

$$x_2 + 10 + x_2 - \left(\frac{x_2 + 5}{2} \right) + 5 = 1$$

$$2x_2 - \frac{x_2}{2} = 14 + \frac{5}{2}$$

$$\frac{5}{2}x_2 = \frac{33}{2}$$

$$x_2 = \frac{33}{5} = x_1 - 10 = 2x_3 - 5$$

$$x_1 = \frac{83}{5}$$

$$x_3 = \frac{58}{10}$$

$$\frac{10}{5}x_3 - \frac{25}{5} = \frac{33}{5}$$

$$\frac{10}{5}x_3 = \frac{58}{5}$$

Решено $(-2, -5, -2, 2, 0)$

$$\delta_1 \begin{array}{cccc|c} 2 & 3 & -5 & 1 & 2 \\ 2 & 3 & -1 & 3 & 8 \\ 6 & 9 & -7 & 7 & 18 \\ 4 & 6 & -12 & 1 & 1 \end{array} \left. \vphantom{\begin{array}{cccc|c}} \right) -2$$

$$\begin{array}{cccc|c} 2 & 3 & -5 & 1 & 2 \\ 2 & 3 & -1 & 3 & 8 \\ 6 & 9 & -7 & 7 & 18 \\ 0 & 0 & -2 & -1 & -3 \end{array} \left. \vphantom{\begin{array}{cccc|c}} \right) -3$$

$$\begin{array}{cccc|c} 2 & 3 & -5 & 1 & 2 \\ 2 & 3 & -1 & 3 & 8 \\ 0 & 0 & -4 & -2 & -6 \\ 0 & 0 & -2 & -1 & -3 \end{array} \left. \vphantom{\begin{array}{cccc|c}} \right) -$$

$$\begin{array}{cccc|c} 2 & 3 & -5 & 1 & 2 \\ 0 & 0 & 4 & 3 & 6 \\ 0 & 0 & -4 & -2 & -6 \\ 0 & 0 & -2 & -1 & -3 \end{array} \left. \vphantom{\begin{array}{cccc|c}} \right) 12+$$

$$\begin{array}{cccc|c} 2 & 3 & -5 & 1 & 2 \\ 0 & 0 & 4 & 3 & 6 \\ 0 & 0 & -4 & -2 & -6 \\ 0 & 0 & 0 & \frac{1}{2} & 0 \end{array}$$

$$x_4 = 0$$

$$4x_5 = 6$$

$$x_3 = \frac{6}{4}$$

$$-4x_3 = -6$$

$$2x_1 + 3x_2 - \frac{30}{4} = 2 \quad 2x_1 + 3x_2 = 3,5$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & \lambda & 1 \\ \lambda & 1 & 1 \end{pmatrix} \begin{array}{c} 1 \\ 1 \\ 1 \end{array} \quad R_2 - R_1, R_3 - \lambda R_1$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & \lambda-1 & 1-\lambda \\ 0 & 1-\lambda & 1-\lambda^2 \end{pmatrix} \begin{array}{c} 1 \\ 0 \\ 1-\lambda \end{array} \quad R_3 + R_2$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & \lambda-1 & 1-\lambda \\ 0 & 0 & 2-\lambda-\lambda^2 \end{pmatrix} \begin{array}{c} 1 \\ 0 \\ -\lambda^2 \end{array}$$

$$1) \quad 2-\lambda-\lambda^2 = 0$$

$$\lambda = 1, -2$$

$$-(\lambda-1)(\lambda+2)$$

$$\lambda, \quad (1-\lambda)(\lambda+2) = 0, \quad \lambda-1 \neq 0$$

$$\lambda \neq 1$$

$$\lambda = -2 \quad R_3 = 10:3$$

$$\Downarrow$$

$$\text{H.P.}$$

$$2) \quad \lambda = 1 \quad R_3 (0 \ 0 \ 1 \ 0)$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{array}{c} 1 \\ 0 \\ 0 \end{array} \Rightarrow x_1 + x_2 + x_3 = 1$$

$$\text{3 y.p.} \Rightarrow$$

$$3) \quad \lambda \neq -2 \quad \lambda \neq 1$$

$$x_2 = p \quad x_3 = q$$

$$1-\lambda \neq 0 \quad R_3 \cdot \frac{1}{1-\lambda} \quad \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad x_1 = 1-p-q$$

$$(1-p-q, p, q)$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & \lambda+2 \end{pmatrix} \begin{array}{c} 1 \\ 0 \\ 1 \end{array} \quad R_3 = x_3 = \frac{1}{\lambda+2}$$

$$R_2 = 1 \quad x_2 = x_3 = \frac{1}{\lambda+2}$$

$$R_1 \Rightarrow x_1 = 1 - \frac{1}{\lambda+2} - \frac{1}{\lambda+2} = \frac{\lambda+2-1-1}{\lambda+2} = \frac{1}{\lambda+2}$$

О.т.

$$1) \lambda = -2 \Rightarrow \text{УР}$$

$$2) \lambda = 1 \Rightarrow (1-p-q, p, q) \neq p, q$$

$$3) \lambda \neq -2, \lambda \neq 1 \quad \left(\frac{1}{\lambda+2}, \frac{1}{\lambda+2}, \frac{1}{\lambda+2} \right); p-q$$

существует

$$v_1, v_2, \dots, v_k \in F^n$$

$$\lambda_1 v_1, \lambda_2 v_2, \dots, \lambda_k v_k \stackrel{x}{=} 0$$

лих комбинация на $v_1, \dots, v_k$

→ с а л з ако $\exists (\lambda_1, \dots, \lambda_k) \neq (0, \dots, 0)$ т.е. x

с а л з не $\exists (\lambda_1, \dots, \lambda_k) \neq (0, \dots, 0)$ т.е.

$$\text{ако } x \Leftrightarrow (\lambda_1, \dots, \lambda_k) = (0, \dots, 0)$$

$\nexists (\lambda_1, \dots, \lambda_k) \neq (0, \dots, 0)$ x не е вярно

$$v \neq 0$$

$$\lambda v = 0 \quad \exists \lambda \neq 0 \text{ т.е.}$$

$$v \neq 0$$

$$\lambda v = 0$$

с а л з ако $\lambda = 0$

$$v_1 = (2, 1, 0, 3, 1)$$

$$v_2 = (-4, -3, -2, 5, -6)$$

$$v_3 = (3, 2, 1, 1, 3)$$

$$v_4 = (0, 0, 0, 1, 0)$$

$$v = (-9, -7, -5, 15, -12)$$

1), 2)

$$1) v = (\lambda_1 v_1 + \lambda_2 v_2 - 1)$$

$$\begin{pmatrix} -9 \\ -7 \\ -5 \\ 15 \\ -12 \end{pmatrix} = \lambda_1 \begin{pmatrix} 2 \\ 1 \\ 0 \\ 3 \\ 1 \end{pmatrix} + \lambda_2 \begin{pmatrix} -4 \\ -3 \\ -2 \\ 5 \\ -6 \end{pmatrix} + \lambda_3 \begin{pmatrix} 3 \\ 2 \\ 1 \\ 1 \\ 3 \end{pmatrix} + \lambda_4 \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} -9 \\ -7 \\ -5 \\ 15 \\ -12 \end{pmatrix} = \begin{pmatrix} 2\lambda_1 \\ \lambda_1 \\ 0 \\ 3\lambda_1 \\ \lambda_1 \end{pmatrix} +$$

$$\begin{pmatrix} -9 \\ -7 \\ -5 \\ 15 \\ -12 \end{pmatrix} = \begin{pmatrix} 2\lambda_1 - 4\lambda_2 + 3\lambda_3 + 0 \\ \lambda_1 - 3\lambda_2 + 2\lambda_3 + 0 \\ 0 - 2\lambda_2 + \lambda_3 + 0 \\ 3\lambda_1 + 5\lambda_2 + \lambda_3 + \lambda_4 \\ \lambda_1 - 6\lambda_2 + 3\lambda_3 + 0 \end{pmatrix}$$

$$\begin{pmatrix} 2 & -4 & 3 & 0 & | & -9 \\ 1 & -3 & 2 & 0 & | & -7 \\ 0 & -2 & 1 & 0 & | & -5 \\ 3 & 5 & 1 & 1 & | & 15 \\ 1 & -6 & 3 & 0 & | & -12 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 2 & -1 & 0 & | & 5 \\ 1 & -3 & 2 & 0 & | & -7 \\ 0 & -2 & 1 & 0 & | & -5 \\ 0 & 14 & -5 & 0 & | & 36 \\ 0 & -3 & 1 & 0 & | & -5 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -3 & 2 & 0 & | & -7 \\ 0 & 2 & -1 & 0 & | & 5 \\ 0 & -2 & 1 & 0 & | & -5 \\ 0 & 14 & -5 & 1 & | & 36 \\ 0 & -3 & 1 & 0 & | & -5 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -3 & 2 & 0 & | & -7 \\ 0 & 2 & -1 & 0 & | & 5 \\ 0 & 0 & -10 & 5 & | & 5 \\ 0 & 0 & 0 & 1 & | & 11 \\ 0 & 0 & 0 & 1 & | & 5 \end{pmatrix} \xrightarrow{R_5 - R_4} \begin{pmatrix} 1 & -3 & 2 & 0 & | & -7 \\ 0 & 2 & -1 & 0 & | & 5 \\ 0 & 0 & -10 & 5 & | & 5 \\ 0 & 0 & 0 & 1 & | & 11 \\ 0 & 0 & -1 & 0 & | & 5 \end{pmatrix}$$

$$\lambda_4 = 11$$

$$\lambda_3 = -5$$

$$\lambda_2 = 0$$

$$\begin{pmatrix} 1 & -3 & 2 & 0 & | & -7 \\ 0 & 2 & -1 & 0 & | & 5 \\ 0 & 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & 0 & | & 0 \\ 0 & 0 & 2 & 1 & | & 1 \end{pmatrix} \xrightarrow{R_5 \cdot \frac{1}{2}} \begin{pmatrix} 1 & -3 & 2 & 0 & | & -7 \\ 0 & 2 & -1 & 0 & | & 5 \\ 0 & 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & 0 & | & 0 \\ 0 & 0 & 1 & \frac{1}{2} & | & \frac{1}{2} \end{pmatrix}$$

$$\begin{pmatrix} 1 & -3 & 2 & 0 & | & -7 \\ 0 & 2 & -1 & 0 & | & 5 \\ 0 & 0 & 1 & \frac{1}{2} & | & \frac{1}{2} \\ 0 & 0 & 0 & 0 & | & 0 \\ 0 & 0 & -1 & 0 & | & 5 \end{pmatrix} \xrightarrow{R_5 + R_3} \begin{pmatrix} 1 & -3 & 2 & 0 & | & -7 \\ 0 & 2 & -1 & 0 & | & 5 \\ 0 & 0 & 1 & \frac{1}{2} & | & \frac{1}{2} \\ 0 & 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & | & \frac{11}{2} \end{pmatrix}$$

$$\lambda_1 + 0 - 10 = -7 \quad \lambda_1 = 3$$

$$2) \lambda_1 v_1 + \lambda_2 v_2 + \lambda_3 v_3 + \lambda_4 v_4 = 0$$

$$\begin{pmatrix} 2 & -4 & 3 & 0 & | & 0 \\ 1 & -3 & 2 & 0 & | & 0 \\ 0 & -2 & 1 & 0 & | & 0 \\ 3 & 5 & 1 & 1 & | & 0 \\ 1 & -6 & 3 & 0 & | & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -3 & 2 & 0 & | & 0 \\ 0 & 2 & -1 & 0 & | & 0 \\ 0 & 0 & 0 & 1 & | & 0 \\ 0 & 0 & -1 & 0 & | & 0 \end{pmatrix}$$

$$4-0, 9-9 \cdot 8$$

$$\rightarrow (0, 0, 0, 0)$$

$\Rightarrow$ como $\text{gr}(\lambda, \lambda_4) = (0, \dots, 0)$

$\Downarrow$
 $v_1, \dots, v_4$ ca LHS

2) $v_1 = (1, 1, \lambda)$ $(2, 3, 4, 5, 6, 2)$

$$v_2 = (1, \lambda, 1)$$

$$v_3 = (\lambda, 1, 1)$$

$$v = (1, 1, 1)$$

$$\begin{array}{ccc|c} 1 & 1 & \lambda & 0 \\ 1 & \lambda & 1 & 0 \\ \lambda & 1 & 1 & 0 \end{array} \rightarrow \Delta$$

$$\begin{pmatrix} 1 & 1 & \lambda & 0 \\ 0 & \lambda-1 & 1-\lambda & 0 \\ 0 & 0 & (1-\lambda)(\lambda+2) & 0 \end{pmatrix}$$

$\Delta(0, 0, 0, 0)$ e pr

1. $\lambda = 1$ $\Rightarrow$ pr e gr $\Rightarrow$ LHS

2. $\lambda \neq 1$ $\Rightarrow$ pr $(0, 0)$ $\Rightarrow$ LHS

3. $\lambda = -2$ $\Rightarrow$ pr $\Rightarrow$ LHS

$$\begin{array}{ccc|c} 1 & 1 & \lambda & 0 \\ 0 & \lambda-1 & 1-\lambda & 0 \\ 0 & 0 & (1-\lambda)(\lambda+2) & 0 \end{array} \begin{array}{l} -1. \lambda = -2 \\ -2. \lambda = 1 - (1-p-q, p, q) \\ -3. \lambda = \left(\frac{1}{\lambda+2}, \frac{1}{\lambda+2}, \frac{1}{\lambda+2} \right) \end{array}$$

4) $\text{gr} \lambda = -2$

$\text{pr} \Rightarrow v$ $\Rightarrow$ pr e gr e pr e gr

5) $\text{gr} \lambda \neq -2, \lambda \neq 1$

pr $\Rightarrow$ pr e gr e pr e gr

6) $\lambda = 1$ $\Rightarrow$ pr e gr

$\Rightarrow v$ e pr e gr e pr e gr

$$v = -1v_1 + 1v_2 + 1v_3$$

$$p_1 = 1, q_1 = 1$$

$$p_2 = 1, q_2 = 1 \quad -v = (-3-i)v_1 + iv_2 + 4v_3$$

7) $\text{gr} v$ $\Rightarrow$ pr e gr e pr e gr e pr e gr

$\text{gr} \lambda = 1$ e $\text{gr} \lambda \neq -2, \lambda \neq 1 \Rightarrow \text{gr} \lambda \neq -2$

$$\begin{pmatrix} 1 & 0 & 0 & | & 0 \\ 0 & 1 & 0 & | & 0 \\ 0 & 0 & 1 & | & 0 \end{pmatrix} \text{ cano } (0,0,0) \Rightarrow \text{ЛНЗ}$$

$$\begin{pmatrix} 1 & 3 & 5 & | & 0 \\ 0 & 1 & 2 & | & 0 \\ 0 & 0 & 1 & | & 0 \end{pmatrix} \Rightarrow \text{ЛНЗ}$$

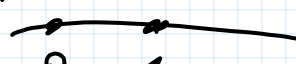
$$\begin{pmatrix} 3 & 8 & x & x & | & 0 \\ 0 & 2 & x & 0 & | & 0 \\ 0 & 0 & 0 & 0 & | & 0 \end{pmatrix} \Rightarrow \text{ЛЗ}$$

$$\begin{pmatrix} i & 0 & 2i & 0 \\ 0 & i & 0 & 0 \\ 3 & 0 & 0 & 0 \end{pmatrix} \Rightarrow \text{ЛНЗ}$$

4 Ракт. МЛНЗП

$$\begin{pmatrix} 2 & -4 & 3 & 0 & | & -9 \\ 1 & -3 & 2 & 0 & | & -7 \\ 0 & -2 & 1 & 0 & | & -5 \\ 3 & 5 & 1 & 1 & | & 15 \\ 1 & 6 & 3 & 0 & | & 12 \end{pmatrix}$$

Пример 1



$$V = \mathbb{R}, F = \mathbb{R}$$

$$v \in \mathbb{R} \Rightarrow \exists \lambda \in F \quad v = \lambda \cdot 1$$

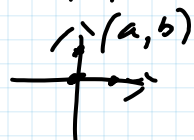
$$v = \lambda_1 v_1 + \lambda_2 v_2 + \lambda_3 v_3 + \lambda_4 v_4$$

или координаты $\lambda_1 v_1 + \dots + \lambda_n v_n$
 ЛЗ ЛНЗ $\lambda_1 + \lambda_2 + \dots + \lambda_n = 0$

или координаты $(v_1, \dots, v_n) = \{ \lambda_1 v_1 + \dots + \lambda_n v_n \}$
 ЛЗ ЛНЗ $\lambda = 0$
 или координаты $(v_1, \dots, v_n) = \{ \lambda_1 v_1 + \dots + \lambda_n v_n \}$
 ЛЗ ЛНЗ $\lambda = 0$
 или координаты $(v_1, \dots, v_n) = \{ \lambda_1 v_1 + \dots + \lambda_n v_n \}$
 ЛЗ ЛНЗ $\lambda = 0$

Пример 2

$$V = \mathbb{R}^2, F = \mathbb{R}$$



$v \in V, \exists \mu_1, \mu_2 \in F$
 $v = \mu_1 v_1 + \mu_2 v_2$
 $(a, b) \in V \Rightarrow \exists \lambda_1, \lambda_2 \in F$
 $(a, b) = \lambda_1 (1, 0) + \lambda_2 (0, 1)$

Ap 3 $V = \mathbb{R}^3$ $f = \mathbb{R}$

$$(0,0,1) | 0,1,0 | 1,0,0 \rightarrow 6 \text{ ug } / R^3$$

ΓP^1 g_{CK} z $1 = 143$

$$V_{123} \subseteq \mathbb{R}^3 \setminus \{0\}$$

$n_p = 2 \quad \left(\begin{array}{c|c} 0 & 1 \\ \hline 1 & 0 \end{array} \right) \cdot \text{cano } Q_{\text{za p-e}}$

$$\left(\begin{array}{c|c} 1 & v_1 \\ \vdots & \vdots \\ v_k & v_k \\ \vdots & \vdots \\ 1 & 1 \end{array} \middle| \begin{array}{c} 0 \\ \vdots \\ 0 \end{array} \right) \begin{array}{l} - \text{ЛЗ} \text{ р-с } \exists (\lambda_1, \lambda_k) \neq (0, \dots, 0)_{\text{р-с}} \\ - \text{ЛНЗ} \text{ с } \lambda_1, \dots, \lambda_k = (0, \dots, 0) \end{array}$$

$$1 \in \mathcal{A} \setminus \{0, \dots, 0\}$$

$\lambda \in \mathcal{P}(\mathcal{C})$

Розумності - брешах вказує в галузі науки

$$\begin{array}{l} \left(\begin{array}{cccc|c} -4 & 3 & 0 & -8 \\ -3 & 2 & 0 & -7 \\ -2 & 1 & 0 & -5 \\ 5 & 1 & 1 & 15 \\ -4 & 3 & 0 & 12 \end{array} \right) \begin{array}{l} \cdot 2 \\ \cdot 3 \\ \cdot 2 \\ \cdot 5 \\ \cdot 1 \end{array} \\ \left(\begin{array}{cccc|c} -2 & -19 & 0 & -3 & -57 \\ -5 & -13 & 0 & -2 & -37 \\ -3 & -7 & 0 & -1 & -20 \\ -3 & 5 & 1 & 15 \\ -8 & -9 & 0 & -3 \end{array} \right) \begin{array}{l} \cdot (-3) \\ \cdot (-2) \\ \cdot (-2) \\ \cdot 1 \\ \cdot (-57) \end{array} \\ \left(\begin{array}{cccc|c} 2 & 2 & 0 & 0 & 6 \\ 1 & 1 & 0 & 0 & 3 \\ 0 & -2 & 1 & 0 & -5 \\ -3 & -7 & 0 & -1 & -20 \\ 1 & 12 & 0 & 0 & 27 \end{array} \right) \begin{array}{l} \cdot (-1) \\ \cdot 3 \\ \cdot 3 \\ \cdot 3 \\ \cdot 3 \end{array} \end{array}$$

$$\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 3 \\ 0 & -2 & 1 & 0 & -5 \\ 0 & -4 & 0 & 2 & -11 \\ 0 & 1 & 0 & 0 & 0^{-1} \end{array}$$

$$R_4 = \lambda_2 = 0$$

$$R_1 = \lambda_1 = 3$$

$$R_2 = \lambda_3 = -5$$

$$R_3 = 1_1$$

Геране саро релл $(3, 0, -5, 11)$

1. $\left(\begin{smallmatrix} 1 \\ \vdots \\ v_k \end{smallmatrix} \middle| \begin{smallmatrix} 1 \\ \vdots \\ 1 \end{smallmatrix} \right)$ чн рел $\Rightarrow V$ е лнх конб

$$V = 3v_1 + 0v_2 - 5v_3 + 11v_4$$

2. $\left(\begin{smallmatrix} 1 \\ \vdots \\ v_k \end{smallmatrix} \middle| \begin{smallmatrix} 0 \\ \vdots \\ 0 \end{smallmatrix} \right)$ чн саро $(0, 0)$ за рел - ЛНЗ

3. нел

4. Дан $v_1, v_2, v_3, v_4, v_5 = V$ са сн снхс на $\mathbb{R}^5$
нел, нел са ЛНЗ

МЛНЗП - вектор нзменду
Първнмалнте конб са ЛНЗ н
са макснм, мнотго тнкнва

Ранг - брн векторн в МЛНЗП

Бннху ч урел - $\exists$ нелуелс
рел

нелуелс v_5 е лнх конб ан v_1, v_2

$\bar{D}_r$ е определн рангът на нелкун. от в-ри

$\{v_1, v_2, v_3, v_4, v_5 = V\}$ н за се нелерн нелкун МЛНЗП
(Дан. - векторн)

r -ч МЛНЗП $\{v_1, v_2, v_3, v_4\}$

нелуелс нел промел р- нелкун слн / нел промел ранг

$\begin{pmatrix} 1 & 1 & 1 \\ 1 & \lambda & 1 \\ 1 & 1 & 1 \end{pmatrix}$ а) D_n е определена матрица на v_1, v_2, v_3 в базиса $\{v_1, v_2, v_3\}$ от λ
 б) да се намери характеристичният полином

$$+ C \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & \lambda & 1 & 0 \\ 0 & 1-\lambda & 1-\lambda & 0 \end{array} \right) \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & \lambda & 1-\lambda & 0 \\ 0 & 0 & -\lambda^2 + \lambda + 2 & 0 \end{array} \right) D=0 \quad \lambda_{1,2} = \frac{1 \pm 5}{-2} \quad \begin{array}{l} \lambda_1 = -2 \\ \lambda_2 = 1 \end{array}$$

$$\text{г) } \lambda = -2$$

$$-\lambda - \lambda^2 - \lambda + 2 = -(\lambda + 2)(\lambda - 1)$$

$$R_3(0 \ 0 \ 0 \ | \ 0)$$

$$\left(\begin{array}{ccc|c} 1 & 1 & -2 & 0 \\ 0 & -3 & 3 & 0 \end{array} \right), \quad \left(\begin{array}{ccc|c} 1 & 1 & -2 & 0 \\ 0 & 1 & -1 & 0 \end{array} \right) \quad \begin{array}{l} 3 \text{ уравнения} \\ \text{ранг } = 2 \end{array}$$

$$\text{Реш. } x_3 = p \quad R_2 \Rightarrow x_2 = p$$

$$R_1 \Rightarrow x_1 = 2p - p = p$$

характеристич. е $\{v_1, v_2\}$ — базис и макс. бр. вект.
 и $\{v_1, v_3\}$
 и $\{v_2, v_3\}$

$$2 \text{ сл } \lambda = 1 \quad \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \quad (1 \ 1 \ 1 \ | \ 0)$$

Полагаме 2 уравн.

— ранг е 1

характеристич. — v_1
 и v_3
 и v_2

$$3 \text{ ca } \lambda \neq -2 \quad \lambda \neq 1$$

$$r=3 \quad \text{MLU37} \{v_1, v_2, v_3\}$$

3a) a)
$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & \lambda-6 & 0 & 0 & 0 \\ 0 & 0 & \mu+\lambda & 0 & 0 \\ 0 & 0 & 0 & \lambda-6 & 0 \\ 0 & 0 & 0 & 0 & \lambda-6 \end{pmatrix}$$

1 ca $\lambda-6=0 \quad \lambda=6 \quad \mu+\lambda \neq 0 \quad \mu \neq -1$
 $\mu \neq -6 \quad r=2$

2 ca $\lambda-6=0 \quad \mu+\lambda=0$
 $\lambda=6 \quad \mu=-6 \quad r=1$

4 ca $\mu+\lambda \neq 0 \quad \lambda-6 \neq 0$
 $\mu \neq -1 \quad \lambda \neq 6$
 $r=5$

3 ca $\lambda-6 \neq 0 \quad \mu+\lambda=0$
 $\lambda \neq 6 \quad \mu = -\lambda$
 $r=4$

b)
$$\begin{pmatrix} 1 & 2 & 3 & \dots & n \\ 0 & \lambda-6 & 0 & 0 & 0 \\ 0 & 0 & \mu+\lambda & 0 & 0 \\ 0 & 0 & 0 & \lambda-6 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

1 ca $\lambda-6=0 \quad \mu+\lambda \neq 0$
 $\lambda=6 \quad \mu \neq -1$
 $r=2$

2 ca $\mu+\lambda=0 \quad \lambda-6=0$
 $\mu=-1 \quad \lambda=6$
 $r=1$

3 ca $\lambda-6 \neq 0 \quad \mu+\lambda=0$
 $\lambda \neq 6 \quad \mu = -1$
 $r=n-1$

4 ca $\lambda-6 \neq 0 \quad \mu+\lambda \neq 0$
 $\lambda \neq 6 \quad \mu \neq -1$
 $r=n$

c)
$$\begin{pmatrix} 1 & 2 & 3 & \dots & n \\ \lambda-1 & \lambda-6 & 0 & 0 & 0 \\ \lambda-2 & 0 & \mu+\lambda & 0 & 0 \\ \lambda-3 & 0 & 0 & \lambda-6 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

$n \times n$

1 ca $\lambda-6 \neq 0 \quad \mu+\lambda \neq 0$

$$C_i = C_1 - \frac{\lambda-1}{\lambda-6} C_2 - \frac{\lambda-2}{\mu+\lambda} C_3 - \frac{\lambda-3}{\lambda-6} C_4 - \dots - \frac{\lambda-(n-1)}{\lambda-6} C_n$$

$$\begin{pmatrix} 0 & 2 & 3 & 4 & \dots & n \\ 0 & \lambda-6 & 0 & 0 & 0 & 0 \\ 0 & 0 & \mu+\lambda & 0 & 0 & 0 \\ 0 & 0 & 0 & \lambda-6 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \end{pmatrix}$$

$\oplus = 1 - \frac{\lambda-1}{\lambda-6} \cdot 2 - \frac{\lambda-2}{\mu+\lambda} \cdot 3 - \frac{\lambda-3}{\lambda-6} \cdot 4 - \dots - \frac{\lambda-(n-1)}{\lambda-6} \cdot n$

2. 7 ca $\lambda=2 \quad \oplus=0 \quad r=n-1$

$$1 \ 2 \times 1 \oplus \neq 0 \quad r=4$$

$$2 \times 1 \quad \lambda - 6 = 0 \quad \mu + \lambda \neq 0$$

$$\lambda = 6 \quad \mu \neq -1$$

$$C_1' = C_1 - \frac{\lambda - 2}{\mu + \lambda} C_3$$

2, 4, 5 ... are proportional

$$2 \leq r \leq 3$$

$$r = 3$$

$$\begin{pmatrix} 0 & 1 & 1 \\ 0 & 6 & 0 \\ 0 & 0 & \mu + \lambda \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & \dots & \dots & \dots & 4 \\ \lambda - 1 & \lambda - 6 & 0 & \dots & \dots & 0 \\ 0 & 0 & \mu + \lambda & \dots & \dots & 0 \\ \lambda - 1 & 0 & 0 & \lambda - 6 & \dots & \dots \\ \lambda - 1 & 0 & 0 & \dots & \dots & 0 \\ \lambda - 1 & \dots & \dots & \dots & \lambda - 6 & \dots \end{pmatrix}$$

$$3 \times 1 \quad \lambda - 6 \neq 0 \quad \mu + \lambda = 0 \quad \text{we have} \quad \lambda - 6 \neq 0 \neq 0$$

$$\begin{pmatrix} 1 & 2 & 3 & \dots & \dots & 4 \\ \lambda - 1 & 0 & 0 & \dots & \dots & 0 \\ \lambda - 2 & 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \lambda - (n-1) & 0 & \dots & \dots & \dots & 0 \end{pmatrix}$$

2, 3, ... are proportional

1, 2 are linearly independent $\Rightarrow r = 2$

3.1

$$\begin{pmatrix} 1 & 2 & 3 & \dots & \dots & 4 \\ 0 & \lambda - 6 & \dots & \dots & \dots & 0 \\ \lambda - 2 & 0 & \mu + \lambda & \dots & \dots & 0 \\ 0 & 0 & 0 & \lambda - 6 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \dots & \dots & \dots & \lambda - 6 & \dots \end{pmatrix}$$

$$(\lambda) = 0$$

$$r = 4 - 1 = 3$$

$$R_1' = R_1 - \frac{2}{\lambda - 6} R_2 - \frac{3}{\lambda - 6} R_3 - \dots - \frac{4}{\lambda - 6} R_n$$

$$\begin{pmatrix} 1 & 0 & 3 & 0 & \dots & 0 \\ 0 & \lambda - 6 & 0 & 0 & \dots & 0 \\ \lambda - 2 & 0 & \mu + \lambda & 0 & \dots & 0 \\ 0 & 0 & 0 & \lambda - 6 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \dots & \dots & \dots & \lambda - 6 & \dots \end{pmatrix}$$

3.1 $\lambda = 2$ - 7 rows and 3 rows are proportional

$$r = 4 - 1$$

3.2 $\lambda = 2 \Rightarrow r = 4$ bbb

2. В а системе рангът е зависим от α и β

$$\begin{pmatrix} 1 & -1 & 2 & 3 & 1 & 4 \\ 1 & 3 & 4 & 7 & -3 & 4 \\ 1 & 0 & 3 & 2 & 0 & 4 \\ 3 & -3 & 6 & \alpha-\beta+9 & \beta & 13 \\ 1 & 1 & -1 & 6 & \beta & \beta+5 \end{pmatrix} \begin{array}{l} \xrightarrow{-1} \\ \xrightarrow{-1} \\ \xrightarrow{-1} \\ \xrightarrow{-3} \\ \xrightarrow{-1} \end{array}$$

$$r \leq 5$$

$$\begin{array}{cccccc} 1 & -1 & 2 & 3 & 1 & 4 \\ 0 & 4 & 2 & -2 & -4 & 0 \\ 0 & 1 & 1 & -1 & -1 & 0 \\ 0 & 0 & 0 & \alpha-\beta & 0 & 1 \\ 0 & 2 & -3 & 3 & \beta-1 & \beta+1 \end{array} \begin{array}{l} \\ \xrightarrow{-4} \\ \\ \xrightarrow{-2} \\ \end{array} \begin{array}{cccccc} 1 & -1 & 2 & 3 & 1 & 4 \\ 0 & 1 & 1 & -1 & -1 & 0 \\ 0 & 0 & -2 & 2 & 0 & 0 \cdot 2 \\ 0 & 0 & 0 & \alpha-\beta & 0 & 1 \\ 0 & 0 & -5 & 5 & \beta+1 & \beta+1 \end{array}$$

$$\begin{array}{cccccc} 1 & -1 & 2 & 3 & 1 & 4 \\ 0 & 1 & 1 & -1 & -1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & \alpha-\beta & 0 & 1 \\ 0 & 0 & -5 & 5 & \beta+1 & \beta+1 \end{array} \begin{array}{l} \\ \\ \xrightarrow{+5} \\ \\ \end{array} \begin{array}{cccccc} 1 & -1 & 2 & 3 & 1 & 4 \\ 0 & 1 & 1 & -1 & -1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & \alpha-\beta & 0 & 1 \\ 0 & 0 & 0 & 0 & \beta+1 & \beta+1 \end{array} \xrightarrow{+}$$

$$\begin{pmatrix} 1 & 0 & 3 & 2 & 0 & 4 \\ 0 & 1 & 1 & -1 & -1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & \alpha-\beta & 0 & 1 \\ 0 & 0 & 0 & 0 & \beta+1 & \beta+1 \end{pmatrix} \begin{array}{l} \\ \\ \\ \\ \end{array}$$

1. $\alpha - \beta = 0 \quad \beta + 1 = 0$
 $\alpha = \beta \quad \beta = -1$
 $\Rightarrow r = 4$

2. $\beta + 1 = 0 \quad \alpha - \beta \neq 0$
 $\beta = -1 \quad \alpha \neq \beta$
 $\Rightarrow r = 4$

$$\begin{array}{ll} 3_{cl} & \beta + 1 \neq 0 \\ & \beta \neq -1 \end{array} \quad \begin{array}{l} \alpha - \beta = 0 \\ \alpha = \beta + 1 \end{array} \quad \begin{array}{ll} 4_{cl} & \beta + 1 \neq 0 \\ & \beta \neq 1 \end{array} \quad \begin{array}{l} \alpha - \beta \neq 0 \\ \alpha \neq \beta \end{array}$$

$$\Rightarrow r = 5$$

$$r = 5$$

Други? конект

2 Гаус

3 F⁴ ми контрове. из из

и рачт ММЗП базис

5 Хомогенна система (букс на ХСЛУ - ФСР)

а) Да се намери базис на пр-вото от р-3 на:

$$\begin{array}{ccccc|c} 2 & 9 & 5 & 2 & 1 & 0 \\ 2 & 7 & 3 & -2 & -12 & 0 \\ 3 & 12 & 6 & 0 & -8 & 0 \\ 3 & 13 & 7 & 2 & -2 & 0 \end{array} \begin{array}{l} (-1) \\ 0 \cdot \frac{1}{2} \\ 0 \cdot \frac{1}{2} \\ 0 \end{array} \begin{array}{l} \\ \\ -\frac{3}{2} \\ -\frac{3}{2} \end{array}$$

ако грешно

$$\begin{array}{ccccc|c} 2 & 9 & 5 & 2 & 1 & 0 \\ 0 & -2 & -2 & -4 & -13 & 0 \\ 0 & -\frac{3}{2} & \frac{3}{2} & -3 & -9\frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & -\frac{1}{2} & -1 & -3\frac{1}{2} & 0 \end{array} \begin{array}{l} \\ \\ \\ \end{array} \begin{array}{l} \\ -4 \\ -3 \end{array}$$

$$\begin{array}{ccccc|c} 2 & 9 & 5 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 3 & 0 & 1 & 0 \\ 0 & -\frac{1}{2} & -\frac{1}{2} & -1 & -3\frac{1}{2} & 0 \end{array} \begin{array}{l} \\ \\ \\ 0 \cdot 2 \end{array} \begin{array}{l} \\ -1 \\ 0 \\ 0 \end{array}$$

$$\begin{array}{ccccc|c} 2 & 9 & 5 & 2 & 1 & 0 \\ 0 & -1 & -1 & -2 & -7 & 0 \\ 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{array} \begin{array}{l} / 2 \\ \\ 0 \\ 0 \end{array} \begin{array}{l} \\ \\ 13 \\ 0 \end{array}$$

$$\begin{array}{ccccc|c} 1 & 4,5 & 2,5 & 1 & 0,5 & 0 \\ 0 & -1 & -1 & -2 & -7 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{array} \begin{array}{l} \\ +4,5 \\ \\ \end{array}$$

$$\begin{array}{ccccc|c} 1 & 0 & -2 & -8 & -3 & 0 \\ 0 & 1 & 1 & 2 & 7 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{array}$$

реамо

$$\begin{array}{ccccc|c} 0 & 0 & 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & -1 & -2 & -6 & 0 \\ 3 & 13 & 7 & 2 & -2 & 0 \end{array}$$

$$x_5 = 0 \quad x_7 = p \quad x_4 = q$$

$$0 - x_2 - x_3 - 2x_4 + 6 \cdot 0 = 0$$

$$-x_2 - p - 2q = 0 \quad x_2 = -(p + 2q)$$

$$3x_1 + 13x_2 + 7x_3 + 2x_4 - 2 \cdot 0 = 0$$

$$3x_1 = +13(p + 2q) - 7p - 2q$$

"
"

$$x_1 = 2p + 8q$$

решение $v = (2p + 8q, -(p + 2q), p, q, 0)$

$$U = \{ 2p + 8q, -(p + 2q), p, q, 0 \mid p, q \in \mathbb{C} \}$$

$$v = p \underbrace{(2, -1, 1, 0, 0)}_{b_1} + q \underbrace{(8, -2, 0, 1, 0)}_{b_2}$$

b_1, b_2 са базис на $U(\phi \subset P)$

8) Да се намери ХСЛУ на това пр-во от р-я дъвопаз

и да се намери базис на V $\hookrightarrow V = \langle v_1, v_2, v_3, v_4 \rangle$

базис на V е $\{ (0, 0, 0, 0, 1), (0, -1, -1, -2, -6), (3, 13, 7, 2, -2) \}$

7 стълба

$$\begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} \left(\begin{array}{ccc|c} & & & 0 \\ & & & 0 \\ & & & 0 \\ & & & 0 \end{array} \right)$$

ФСР: $(2, -1, 1, 0, 0)$
 $(6, -2, 0, 1, 0)$

$$\left(\begin{array}{ccccc|c} 2 & -1 & 1 & 0 & 0 & 0 \\ 6 & -2 & 0 & 1 & 0 & 0 \end{array} \right) \rightarrow \text{пробото}$$

на р-я на тгга
 $x \in V$

Схематично

$$0.2 - 1. - 1 - 1.7 - 2.0 - 6.0 = 0$$

$$0 + 1 - 1 - 0 - 0 = 0$$

Step 3

$$\left(\begin{array}{ccc|c} 1 & 1 & i & 0 \\ 0 & 1-i & 0 & 0 \\ 2-i & 0 & 0 & 0 \end{array} \right) R_3 = \frac{1}{2-i} =$$

$$\frac{1}{2-i} = \frac{1}{2-i} \cdot \frac{2+i}{2+i} = \frac{2+i}{2^2 + 1^2} = \frac{2+i}{5}$$

$$\left(\begin{array}{ccc|c} i & 1 & i & 0 \\ 0 & 1-i & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \xrightarrow{R_2 \cdot \frac{1}{1-i}} \left(\begin{array}{ccc|c} i & 1 & i & 0 \\ 0 & 1 & i & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \xrightarrow{-i} \left(\begin{array}{ccc|c} 0 & 0 & i & 0 \\ 0 & 1 & i & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \xrightarrow{\cdot \frac{1}{i}} \left(\begin{array}{ccc|c} 0 & 0 & 1 & 0 \\ 0 & 1 & i & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$$\begin{matrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{matrix}$$

$$x_1, x_2, x_3 = 0, 0, 0$$

$$\begin{array}{ccccc|c} 2 & 9 & 5 & 2 & 1 & 0 \\ 2 & 7 & 3 & -2 & -12 & 0 \\ 3 & 12 & 6 & 0 & -8 & 0 \\ 3 & 13 & 7 & 2 & -2 & 0 \end{array}$$

a) ? δ на 4 (ФЧР)

б) δ на V и ? ХСЛЧ метод прл,

в) ? δ на 4 NV

р-я
слага с
V

б) δ на V и $V \in V = \{(0, 0, 0, 0, 1), (0, -1, -1, -2, -6), (3, 13, 7, 2, -2)\}$

δ на V
? ХСЛЧ

$$\left(\begin{array}{ccccc|c} 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & -1 & -2 & -6 & 0 \\ 3 & 13 & 7 & 2 & -2 & 0 \end{array} \right)$$

$$\left(\begin{array}{ccccc|c} 2 & -1 & ? & 0 & 0 & 0 \\ 8 & -2 & 0 & ? & 0 & 0 \end{array} \right) \quad \text{тогда } x_1 = p \quad x_2 = q \quad x_5 = r$$

$$R_2 = 7 \quad x_4 = -8x_1 + 2x_2 \quad x_1 = -8p + 2q$$

$$x_5 = -2x_1 + x_2 = -2(-8p + 2q) + q = 16p - 4q + q = 16p - 3q$$

$$(p, q, -2p + q, -8p + 2q, r)$$

$$\phi \subset \mathbb{P} \quad p=1, q=0, r=0 \rightarrow (1, 0, -2, -8, 0)$$

$$p=0, q=1, r=0 \rightarrow (0, 1, -1, 2, 0)$$

$$p=0, q=0, r=1 \rightarrow (0, 0, 0, 0, 1)$$

Ник и 2 ???

$$1 \quad 4. \quad \left(\begin{array}{ccccc|c} 1 & 2 & 3 & 4 & 5 & 0 \\ 2 & 1 & 2 & -1 & 2 & 0 \\ 0 & 1 & 2 & 3 & 1 & 0 \\ -1 & 2 & 5 & 8 & 4 & 0 \end{array} \right) \begin{array}{l} \downarrow \cdot 2 \\ \downarrow +1 \end{array}$$

$$\left(\begin{array}{ccccc|c} 1 & 2 & 3 & 4 & 5 & 0 \\ 0 & -3 & -4 & -9 & -8 & 0 \\ 0 & 1 & 2 & 3 & 1 & 0 \\ 0 & 4 & 6 & 12 & 8 & 0 \end{array} \right) \begin{array}{l} -2 \\ +3 \\ -4 \end{array} \left(\begin{array}{ccccc|c} 1 & 0 & -1 & -2 & 3 & 0 \\ 0 & 0 & 2 & 0 & -5 & 0 \\ 0 & 1 & 2 & 3 & 1 & 0 \\ 0 & 0 & -2 & 0 & 5 & 0 \end{array} \right)$$

$$\left(\begin{array}{ccccc|c} 1 & 0 & -1 & -2 & 3 & 0 \\ 0 & 1 & 2 & 3 & 1 & 0 \\ 0 & 0 & 2 & 0 & -5 & 0 \\ 0 & 0 & -2 & 0 & 5 & 0 \end{array} \right) + \left(\begin{array}{ccccc|c} 1 & 0 & -1 & -2 & 3 & 0 \\ 0 & 1 & 2 & 3 & 1 & 0 \\ 0 & 0 & 2 & 0 & -5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right) -7$$

$$\left(\begin{array}{ccccc|c} 1 & 0 & -1 & -2 & 3 & 0 \\ 0 & 1 & 0 & 3 & 6 & 0 \\ 0 & 0 & 2 & 0 & -5 & 0 \end{array} \right) \quad \begin{array}{l} \zeta - 3 \text{ и } \delta \text{ — свободные параметры} \\ \text{положим } x_5 = 2\rho, x_4 = q \\ R_3 \Rightarrow 2x_3 = 5 \cdot 2\rho \Rightarrow x_3 = 5\rho \end{array}$$

$$R_2 \quad x_2 = -3 \cdot x_4 - 6x_5 \quad x_5 = 2\rho$$

$$x_2 = -3q - 12\rho$$

$$R_1 \quad x_1 = x_3 + 2x_4 - 3x_5 = 5\rho + 2q - 3 \cdot 2\rho = 2q - \rho$$

$$(2q - \rho, -3q - 12\rho, 5\rho, q, 2\rho)$$

$$\phi \subset \rho \cdot \begin{pmatrix} -1, -12, 5, 0, 2 \end{pmatrix} \quad \begin{array}{l} p=1, q=0 \\ p=0, q=1 \end{array}$$

$$(2, -3, 0, 1, 0)$$

б) ?δ на V , когато $V = (v_1, v_2, v_3)$

$$v_1 = (2, -3, 0, 7, 0), v_2 = (3, 9, -5, 7, -2), v_3 = (5, 6, -5, 2, -2)$$

$$\left(\begin{array}{ccc|c} 2 & 3 & 5 & 0 \\ -3 & 9 & 6 & 0 \\ 0 & -5 & -5 & 0 \\ 7 & 7 & 2 & 0 \\ 0 & -2 & -2 & 0 \end{array} \right) \xrightarrow{2-12} \left(\begin{array}{ccc|c} 2 & 3 & 5 & 0 \\ 0 & 12 & 12 & 0 \\ 0 & 7 & 1 & 0 \\ 0 & -2 & -2 & 0 \end{array} \right) \xrightarrow{2-12} \left(\begin{array}{ccc|c} 2 & 3 & 5 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 7 & 1 & 0 \\ 0 & -2 & -2 & 0 \end{array} \right)$$

Показваме ?
неизв

v_1, v_2 - ЛНЗ $\Rightarrow$ базис на V
и са ЛЛНЗ

$$\left(\begin{array}{ccccc|c} 2 & -3 & 0 & 7 & 0 & 0 \\ 3 & 9 & -5 & 7 & -2 & 0 \\ 5 & 6 & -5 & 2 & -2 & 0 \end{array} \right) \xrightarrow{\times 2} \left(\begin{array}{ccccc|c} 2 & -3 & 0 & 7 & 0 & 0 \\ 6 & 18 & -10 & 14 & -4 & 0 \\ 10 & 12 & -10 & 4 & -4 & 0 \end{array} \right) \xrightarrow{2-3} \left(\begin{array}{ccccc|c} 2 & -3 & 0 & 7 & 0 & 0 \\ 0 & 27 & -10 & -7 & -4 & 0 \\ 0 & 27 & -10 & -7 & -4 & 0 \end{array} \right)$$

$$\left(\begin{array}{ccccc|c} 2 & -3 & 0 & 7 & 0 & 0 \\ 0 & 27 & -10 & -7 & -4 & 0 \\ 0 & 27 & -10 & -7 & -4 & 0 \end{array} \right) \xrightarrow{2-3} \left(\begin{array}{ccccc|c} 2 & -3 & 0 & 7 & 0 & 0 \\ 0 & 27 & -10 & -7 & -4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right) \Rightarrow \tilde{v}_1, \tilde{v}_2 \text{ са базис на } V, \text{ защото ЛНЗ и са максимален брой}$$

в) ?δ на $U(V)$ - из които пр.во от 1-я на ХСЛУ

1 пр.во от 1-я на ХСЛУ
по усл. е дадено така

2 пр.во $\tilde{v}$ неизв $\Rightarrow 3$ пар.

$$2 \text{ пр.во } 27x_2 - 10x_3 - x_4 - 4x_5 = 0$$

$$x_2 = p \quad x_3 = q \quad x_5 = r$$

$$27p - 10q - 4r = x_4$$

$$1 \text{ пр.во } 2x_1 - 3x_2 + x_4 = 0$$

$$2x_1 = 3p - 27p + 10q + 4r$$

$$x_1 = -12p + 5q + 2r$$

$$(-12p + 5q + 2r, p, q, 27p - 10q - 4r, r)$$

$$p=1 \quad q=0 \quad r=0$$

$$(-12, 7, 0, 27, 0)$$

$$p=0 \quad q=1 \quad r=0$$

$$(15, 0, 1, -10, 0)$$

$$p=0 \quad q=0 \quad r=1$$

$$(12, 0, 0, -4, 1)$$

$$\begin{cases} 15x_1 + x_2 + 0 + 27x_4 + 0 = 0 \\ -5x_1 + 0 + x_3 - 10 + 0 = 0 \\ -2x_1 + 0 + 0 - 4x_4 + x_5 = 0 \end{cases} \quad \left(\begin{array}{ccccc|c} 15 & 1 & 0 & 27 & 0 & 0 \\ -5 & 0 & 1 & -10 & 0 & 0 \\ -2 & 0 & 0 & -4 & 1 & 0 \end{array} \right)$$

$V \in \text{pr. ker of } p \rightarrow \text{no Tagu}$

$U \cap V$

NP $\xrightarrow{\text{bo}}$ $\xrightarrow{\text{pr. ker of } p}$ $\xrightarrow{\text{no Tagu}}$ $U = \left(\begin{array}{ccccc|c} 1 & 0 & -1 & -2 & 3 & 0 \\ 0 & 1 & 0 & 3 & 6 & 0 \\ 0 & 0 & 2 & 0 & -5 & 0 \end{array} \right)$

$$\left(\begin{array}{ccccc|c} 15 & 1 & 0 & 27 & 0 & 0 \\ -5 & 0 & 1 & -10 & 0 & 0 \\ -2 & 0 & 0 & -4 & 1 & 0 \\ 1 & 0 & -1 & -2 & 3 & 0 \\ 0 & 1 & 0 & 3 & 6 & 0 \\ 0 & 0 & 2 & 0 & -5 & 0 \end{array} \right)$$

$\sim \dots$

$$\sim \left(\begin{array}{ccccc|c} -2 & 0 & 0 & 4 & -1 & 0 \\ 5 & 0 & 1 & -10 & 0 & 0 \\ -12 & 1 & 0 & 27 & 0 & 0 \end{array} \right)$$

$$x_1 = p \quad x_4 = q$$

$$R_3 \Rightarrow x_2 = 12p - 27q$$

$$R_2 \Rightarrow x_3 = 10q - 5p$$

$$R_1 \Rightarrow x_5 = 4q - 2p$$

$$(x_1, x_2, x_3, x_4, x_5) =$$

$$\rightarrow U \cap V = \left\{ (p, 12p - 27q, 10q - 5p, 2p + 4q) \mid p, q \in \mathbb{C} \right\}$$

$$\text{Basis an } U \cap V \text{ e } p=1, q=0 \quad (1, 12, -5, 0, -2)$$

$$p=0, q=1 \quad (0, -27, 10, 2, 4)$$

$$\begin{pmatrix} 1 & 5 & 0 \\ 0 & i^3 - 1 \end{pmatrix}$$

$$\begin{matrix} i^5 = i \\ i^3 = -i \end{matrix}$$

$$i^4 = \begin{cases} 1, & n = 4k \\ i, & n = 4k+1 \\ -1, & n = 4k+2 \\ -i, & n = 4k+3 \end{cases}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 - i \end{pmatrix} \rightarrow r=2$$

$$\left(\frac{-5i - 5\sqrt{3}}{63} \right)^{251} = w$$

$$z = a + ib$$

$$|z| = \sqrt{a^2 + b^2}$$

$$z = r \left(\frac{\alpha}{r} + i \frac{b}{r} \right) = r(\cos \varphi + i \sin \varphi)$$

$$\frac{5}{63} (-i - \sqrt{3}) = \frac{5}{63} \cdot 2 \cdot \left(-\frac{\sqrt{3}}{2} + (-\frac{1}{2})i \right) : \begin{matrix} a = -\sqrt{3} \\ b = -1 \end{matrix}$$

$$\begin{aligned} \frac{\pi + \pi}{6} &= \frac{2\pi}{6} \\ -\pi + \frac{\pi}{6} &= -\frac{5\pi}{6} \end{aligned}$$

$$= \frac{10}{63} \left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} \right)$$

$$w = \left(\frac{10}{63} \right)^{251} \cdot \left(\cos \left(\frac{7}{6} \cdot 251 \pi \right) + i \sin \left(\frac{7}{6} \cdot 251 \pi \right) \right)$$

$$\frac{7,251}{1757:6}$$

$$(\sqrt{3} + i)^{39}$$

$$\begin{array}{r} 300 \\ 280 \\ 284 \quad 13 \end{array}$$

$$r = \sqrt{\sqrt{3}^2 + 1^2} = \sqrt{3+1} = 2$$

$$\sqrt{3} + i = 2 \cdot \left(\frac{\sqrt{3}}{2} + i \frac{1}{2} \right) = 2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$$

$$2^{39} \left(\cos \frac{\pi \cdot 39}{6} + i \sin \frac{\pi \cdot 39}{6} \right)$$

$$\frac{1\pi + 2k\pi}{2}$$

$$2^{39} \left(\cos \frac{\pi}{2} + 2k\pi + i \sin \left(\frac{\pi}{2} + 2k\pi \right) \right) = r^{39} \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right)$$

$$v_1 = (\lambda, \mu, \lambda, \mu, \lambda)$$

$$v_2 = (-1, 1, 1, 1, 1)$$

$$v_3 = (2, 1, 1, 1, 1)$$

$$v_4 = (1+\mu, 3\lambda, 5-\lambda+2\mu, 0, 1)$$

$$v_5 = (0, 0, 0, 0, 1)$$

$$v = (\lambda, 2\lambda, 3\lambda, 4\lambda, 5\lambda)$$

$$? v \in \langle v_1, \dots, v_5 \rangle$$

$$? v_1, v_5 \in \mathbb{R}^5$$

$$x_1 v_1 + \dots + x_5 v_5 = v$$

$$\begin{pmatrix} \lambda & -1 & 2 & 1+\mu & 0 \\ \mu & 1 & 1 & 3-\lambda & 0 \\ \lambda & 1 & 1 & 5-\lambda+2\mu & 0 \\ \mu & 1 & 1 & 0 & 0 \\ \lambda & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} \lambda \\ 2\lambda \\ 3\lambda \\ 4\lambda \\ 5\lambda \end{pmatrix} \begin{matrix} -2 \\ -1 \\ -1 \\ -1 \\ -1 \end{matrix}$$

$$\begin{pmatrix} \lambda-2\mu & -3 & 0 & 1+\mu & 0 \\ 0 & 0 & 0 & 3-\lambda & 0 \\ \lambda-\mu & 0 & 0 & 5-\lambda+2\mu & 0 \\ \mu & 1 & 1 & 0 & 0 \\ \lambda-\mu & 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} -2\lambda \\ -2\lambda \\ -\lambda \\ 4\lambda \\ \lambda \end{pmatrix} \begin{matrix} -1 \\ -1 \\ -1 \\ -1 \\ -1 \end{matrix}$$

$$\begin{pmatrix} \lambda-2\mu & -3 & 0 & 1+\mu & 0 \\ 0 & 0 & 0 & 3-\lambda & 0 \\ \lambda-\mu & 0 & 0 & 2+2\mu & 0 \\ \mu & 1 & 1 & 0 & 0 \\ \lambda-\mu & 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} -2\lambda \\ -2\lambda \\ -3\lambda \\ 4\lambda \\ \lambda \end{pmatrix}$$

$$1.1 \quad 3-\lambda = 0$$

$$R_2 \Rightarrow (0 \ 0 \ | \ -6) \Rightarrow \text{н.р. (неверная система)}$$

$$2.1 \quad \begin{matrix} 3-\lambda \neq 0 \\ \lambda \neq 3 \end{matrix} \quad R_2 \quad x_4 = \frac{-2\lambda}{3-\lambda}$$

$$2.2 \quad \lambda - \mu \neq 0$$

$$R_3 \quad x_1 = (-3\lambda - 2(1+\mu)x_4) / (\lambda - \mu)$$

$$R_5 \Rightarrow x_5 = \lambda - (\lambda - \mu)x_1 - x_4$$

$$\Rightarrow x_2, x_3 = \dots$$

$$2.2) \lambda - \mu = 0$$

$$\text{Отг } 1) \text{ и.р.} \Rightarrow v \notin (v_1 \dots v_5) \text{ ж.к. } \lambda, \mu$$

$$\exists p \in \mathbb{R} \setminus \{0\} \text{ и } p e = v \text{ и представя по единствен начин като л.к. на } v_1, v_5$$

— полова от 1

$$v = x_1 v_1 + \dots + x_5 v_5 \text{ където } x_1, x_5 \text{ рел. к. } v_1, \dots, v_5$$

$$\begin{array}{cccc|c} 1 & \lambda & \mu & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & \lambda - \mu & 0 & 1 & 0 \\ 0 & 0 & \mu + 2\lambda & 0 & \lambda \end{array}$$

$$\begin{aligned} \text{Ж.к. } \mu - 2\lambda &= 0 \\ \mu &= 2\lambda \\ \lambda &= -\frac{\mu}{2} \end{aligned}$$

$$\begin{aligned} \text{Ж.к. } \lambda \neq 0, \mu \neq 0 \\ R_4 (0000 | \lambda) \rightarrow \lambda \mu \end{aligned}$$

$$\text{Ж.к. } \mu = 0, \lambda = 0$$

$$R_4 (0000 | 0)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right) \text{ — Редукция}$$

$$\begin{aligned} x_2 &= p \\ x_3 &= q \end{aligned}$$

$$R_2 \quad x_4 = 0$$

$$R_1 \Rightarrow x_1 = -\lambda v_1 = 0$$

$$x_1 = 0$$

$$(\odot p, q, 0) \neq p, q$$

* СЛУ

чн р е
известна

чн р. е
известна

! - р е
определена

— чн в о р - я
неопределена